

Practical 10 - Estimating causal effects using observational data 1

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: rotterdam_p10

R packages required: survival, tableone, pec, adjustedCurves, riskRegression, mexhaz

```
library(survival)
library(tableone)
library(pec)
library(adjustedCurves)
library(riskRegression)
library(mexhaz)
```

Introduction

In this practical you will perform analyses to obtain the results shown in the lecture, using the Rotterdam breast cancer data. The Rotterdam data comprises data on 2982 individuals who underwent surgery for primary breast cancer and whose data were recorded in the Rotterdam Tumour Bank. Participants were followed-up for up to 20 years for the outcomes of disease recurrence and death from any cause. In this practical the focus is going to be on the composite outcome of disease recurrence or death without recurrence, and on the first 5 years of follow-up. The data have been modified from the original to apply administrative censoring at 5 years.

The variables we will use are summarised in the table below.

Variable name	Description
pid	Patient identifier
year_cat	Year of surgery. 0: before 1990, 1: 1990 or later
age	Age at surgery
meno	Menopausal status. 0=premenopausal, 1=postmenopausal
size	Tumour size: categorical with levels ≤ 20 mm, 20 – 50mm, > 50mm
grade	Differentiation grade: categorical with levels 2 and 3
enodes	Number of positive lymph nodes, transformed as $\text{enodes} = \exp(-0.12 \cdot \text{nodes})$
pgr	Progesterone receptors (fmol/l)
er	Estrogen receptors (fmol/l)
chemo	Chemotherapy: 0=no, 1=yes
hormon	Hormonal treatment: 0=no, 1=yes
time	Years to the first of recurrence or death without recurrence
outcome	Indicator of composite outcome: 0=censored, 1=event
outcome_cr	Competing events outcome: 0=censored, 1=recurrence, 2=death without recurrence

Aims

The aims of this session are:

- Estimate conditional and marginal survival probabilities at a specific time under different treatment options.
- Estimate causal effects of a treatment in terms of marginal risk differences, risk ratios and restricted mean survival times.
- Estimate marginal survival curves under different treatment options.

Question 1

- Load the data.
- How many events are there and how many used hormone therapy?
- Compare the characteristics of users and non-users of hormone therapy. The `tableone` package is useful for this:

```
table1<-CreateTableOne(
  vars=c("year_cat","age","meno","size","grade","enodes",
        "pgr","er","hormon","chemo"),
  strata="hormon",
```

```

data=roott,
factorVars=c("year_cat","meno","size","grade","hormon","chemo"),
includeNA = FALSE,
test = F,
addOverall = T
)

print(table1,showAllLevels = T)

```

SOLUTIONS

```

#load the data
roott<-read.table("datasets/rotterdam_p10.csv",sep=" ",header=T)

#summaries
dim(roott)#2982 people

```

```
[1] 2982    14
```

```
table(roott$outcome) #1285 events
```

```

  0    1
1697 1285

```

```
table(roott$hormon)#339 users of hormone therapy
```

```

  0    1
2643 339

```

```

#The tableone package is useful for creating summary tables
#such as that in Table 10.2 in the notes

```

```

table1<-CreateTableOne(
  vars=c("year_cat","age","meno","size","grade","enodes",
        "pgr","er","hormon","chemo"),

```

```

strata="hormon",
data=roth,
factorVars=c("year_cat","meno","size","grade","hormon","chemo"),
includeNA = FALSE,
test = F,
addOverall = T
)

print(table1,showAllLevels = T)

```

		Stratified by hormon		
		level Overall	0	1
n		2982	2643	339
year_cat (%)	0	1918 (64.3)	1816 (68.7)	102 (30.1)
	1	1064 (35.7)	827 (31.3)	237 (69.9)
age (mean (SD))		55.06 (12.95)	54.10 (12.98)	62.55 (9.92)
meno (%)	0	1312 (44.0)	1271 (48.1)	41 (12.1)
	1	1670 (56.0)	1372 (51.9)	298 (87.9)
size (%)	<=20	1387 (46.5)	1283 (48.5)	104 (30.7)
	>50	304 (10.2)	241 (9.1)	63 (18.6)
	20-50	1291 (43.3)	1119 (42.3)	172 (50.7)
grade (%)	2	794 (26.6)	735 (27.8)	59 (17.4)
	3	2188 (73.4)	1908 (72.2)	280 (82.6)
enodes (mean (SD))		0.80 (0.26)	0.82 (0.25)	0.57 (0.24)
pgr (mean (SD))		161.83 (291.31)	168.71 (300.34)	108.23 (200.30)
er (mean (SD))		166.59 (272.47)	164.79 (272.56)	180.61 (271.69)
hormon (%)	0	2643 (88.6)	2643 (100.0)	0 (0.0)
	1	339 (11.4)	0 (0.0)	339 (100.0)
chemo (%)	0	2402 (80.5)	2091 (79.1)	311 (91.7)
	1	580 (19.5)	552 (20.9)	28 (8.3)

- Among the 2982 people, there are 1285 events in the first 5 years of follow-up, with the remaining 1697 individuals being censored.
- 339 people used hormone therapy.
- Hormone therapy users tend to be older, more likely to be post-menopausal, tended to have larger tumour size and higher grade, tended to have lower values for pgr but higher values for er, and were less likely to have chemotherapy. Hormone therapy use was also more common in the post-1990 period.

Question 2

- Fit an unadjusted Cox model including only the hormone therapy variable. Interpret the hazard ratio estimate.
- Fit a Cox model with hormone therapy as your main exposure, and conditioning on all other explanatory variables included in the data. The variables to include are: `age`, `meno`, `size`, `grade`, `enodes`, `pgr`, `er`, `year_cat`, `chemo`. We will assume that these are all potential confounders. Interpret the hazard ratio estimate for the hormone therapy variable and compare it with the unadjusted estimate.

SOLUTIONS

```
#Fit unadjusted and adjusted Cox regressions
```

```
cox.unadj<-coxph(Surv(time,outcome)~hormon,data=roth)  
summary(cox.unadj)
```

Call:

```
coxph(formula = Surv(time, outcome) ~ hormon, data = roth)
```

```
n= 2982, number of events= 1285
```

```
      coef exp(coef) se(coef)      z Pr(>|z|)  
hormon 0.23899   1.26996  0.08235  2.902  0.00371 **  
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
      exp(coef) exp(-coef) lower .95 upper .95  
hormon      1.27      0.7874      1.081      1.492
```

```
Concordance= 0.512 (se = 0.005 )
```

```
Likelihood ratio test= 7.94 on 1 df, p=0.005
```

```
Wald test = 8.42 on 1 df, p=0.004
```

```
Score (logrank) test = 8.46 on 1 df, p=0.004
```

```
cox.adj<-coxph(Surv(time,outcome)~hormon+age+meno+size+  
               as.factor(grade)+enodes+pgr+er+year_cat+chemo,  
               data=roth,x=TRUE)  
summary(cox.adj)
```

Call:

```
coxph(formula = Surv(time, outcome) ~ hormon + age + meno + size +  
      as.factor(grade) + enodes + pgr + er + year_cat + chemo,  
      data = rott, x = TRUE)
```

n= 2982, number of events= 1285

	coef	exp(coef)	se(coef)	z	Pr(> z)	
hormon	-0.3154887	0.7294323	0.0935751	-3.372	0.000748	***
age	-0.0044099	0.9955998	0.0037513	-1.176	0.239772	
meno	0.0474792	1.0486244	0.0990546	0.479	0.631708	
size>50	0.5915522	1.8067908	0.0937010	6.313	2.73e-10	***
size20-50	0.3613475	1.4352622	0.0653031	5.533	3.14e-08	***
as.factor(grade)3	0.3485967	1.4170776	0.0717021	4.862	1.16e-06	***
enodes	-1.9166310	0.1471017	0.1062511	-18.039	< 2e-16	***
pgr	-0.0002368	0.9997632	0.0001222	-1.937	0.052706	.
er	-0.0002229	0.9997771	0.0001195	-1.866	0.062056	.
year_cat	-0.1632737	0.8493587	0.0646443	-2.526	0.011546	*
chemo	-0.3300663	0.7188761	0.0813020	-4.060	4.91e-05	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
hormon	0.7294	1.3709	0.6072	0.8763
age	0.9956	1.0044	0.9883	1.0029
meno	1.0486	0.9536	0.8636	1.2733
size>50	1.8068	0.5535	1.5037	2.1710
size20-50	1.4353	0.6967	1.2628	1.6312
as.factor(grade)3	1.4171	0.7057	1.2313	1.6309
enodes	0.1471	6.7980	0.1194	0.1812
pgr	0.9998	1.0002	0.9995	1.0000
er	0.9998	1.0002	0.9995	1.0000
year_cat	0.8494	1.1774	0.7483	0.9641
chemo	0.7189	1.3911	0.6130	0.8431

Concordance= 0.691 (se = 0.007)

Likelihood ratio test= 577 on 11 df, p=<2e-16

Wald test = 652.8 on 11 df, p=<2e-16

Score (logrank) test = 728.8 on 11 df, p=<2e-16

- In the unadjusted model, the HR for hormone therapy is 1.27 (95% CI 1.08 to 1.49). In other words, hormone therapy use is associated with a 27% higher hazard for the composite event of recurrence or death (95% CI 8% to 49%).

- After conditioning on the set of variables listed above, the HR for hormone therapy is 0.73 (95% CI 0.61 to 0.88). That is, hormone therapy use is associated with a lower hazard for the composite event of recurrence or death after conditioning on the potential confounders.

Question 3

In this question we will use the adjusted model fitted in Question 2 to obtain an estimate of the risk of the composite event up to 5 years for each individual in the data under two treatment strategies:

- Start hormone therapy following surgery (and continue as recommended by the doctor) (`hormon=1`),
- Do not start hormone therapy following surgery (`hormon=0`).

Use the code below to do this for (i), and modify it for (ii).

What is the estimated risk of recurrence or death up to 5 years for person 1 in the data if they were to have followed treatment strategies (i) and (ii)?

```
#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth
data.hormon1$hormon<-1

#predicted survival probability at time 5 for each individual using cox.adj
#Using predictSurvProb from pec package
risk.hormon1.time5<-1-predictSurvProb(cox.adj,newdata=data.hormon1,times=5)
```

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```
#---
#create dataset in which hormon is set to 1 for everyone
data.hormon1<-roth
data.hormon1$hormon<-1
#---
#create dataset in which hormon is set to 0 for everyone
data.hormon0<-roth
data.hormon0$hormon<-0
#---
```

```

#predicted survival probability at time 5 for each individual using cox.adj
#Using survfit - a bit slow
# surv.hormon1.time5<-survfit(cox.adj,newdata=data.hormon1)
# risk.hormon1.time5<-1-surv.hormon1.time5$surv[surv.hormon1.time5$time==5,]
#
# surv.hormon0.time5<-survfit(cox.adj,newdata=data.hormon0)
# risk.hormon0.time5<-1-surv.hormon0.time5$surv[surv.hormon0.time5$time==5,]

#---
#predicted risk up to time 5 for each individual using cox.adj
#Using predictSurvProb from pec package - neater
risk.hormon1.time5<-1-predictSurvProb(cox.adj,newdata=data.hormon1,times=5)
risk.hormon0.time5<-1-predictSurvProb(cox.adj,newdata=data.hormon0,times=5)

#Take a look at the estimates for the first 5 people
cbind(risk.hormon1.time5,risk.hormon0.time5)[1:5,]

```

```

      [,1]      [,2]
[1,] 0.1988058 0.2620425
[2,] 0.2889956 0.3734909
[3,] 0.1883930 0.2488624
[4,] 0.3357231 0.4292403
[5,] 0.2133252 0.2803147

```

For an individual with the same characteristics as person 1 in the data, except for their hormone therapy status, the risk of the composite event up to 5 years is estimated to be 20% if they were to take hormone therapy and 26% if they were not to take hormone therapy.

Question 4

In this question we will use the results from the previous question to obtain estimates of the marginal (adjusted) risks under the two treatment strategies.

- Obtain the marginal risk estimates $\Pr(T \leq 5 | do(X = 1))$ and $\Pr(T \leq 5 | do(X = 0))$, where X denotes hormone therapy strategy and T denotes the time of the composite event of interest. This can be done by taking the average of the conditional risk estimates obtained in question 3.
- Obtain estimates of the risk difference $\Pr(T \leq 5 | do(X = 1)) - \Pr(T \leq 5 | do(X = 0))$ and the risk ratio $\Pr(T \leq 5 | do(X = 1)) / \Pr(T \leq 5 | do(X = 0))$.

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```
#mean risk up to time 5, averaging over all individuals
riskmean.hormon1.time5<-mean(risk.hormon1.time5)
riskmean.hormon0.time5<-mean(risk.hormon0.time5)

riskmean.hormon1.time5
```

```
[1] 0.3639846
```

```
riskmean.hormon0.time5
```

```
[1] 0.4480378
```

```
#Risk difference
riskmean.hormon1.time5-riskmean.hormon0.time5
```

```
[1] -0.08405317
```

```
#Risk ratio
riskmean.hormon1.time5/riskmean.hormon0.time5
```

```
[1] 0.8123971
```

- If everyone had taken hormone therapy the expected marginal (population average) probability of recurrence or death up to 5 years is 36.4%. If everyone had not taken hormone therapy the expected marginal (population average) probability of recurrence or death up to 5 years is 44.8%.
- The marginal risk difference (hormone therapy versus no hormone therapy) is -8.4%. If everyone had taken hormone therapy the risk of recurrence or death within 5 years would have been 8.4 percentage points lower than if everyone had not taken hormone therapy.
- The marginal risk ratio is 0.81. If everyone had taken hormone therapy the risk of recurrence or death within 5 years would have been 19% lower than if everyone had not taken hormone therapy.

Question 5

- In this question we will use a ready-made function `adjustedsurv` from the `adjustedCurves` package to estimate the marginal (adjusted) risks differences and risk ratios, alongside confidence intervals. Take a look at the code below to do this.
- The estimates can be obtained using `adjsurv$sate_object`. Find the marginal risks and the marginal risk difference and risk ratio, and their 95% confidence intervals

```
#It turns out that `hormon` needs to be a factor variable for this
rott$hormonfac<-as.factor(rott$hormonfac)

#Refit Cox model with hormon as factor
#Also need to use x=TRUE here
cox.adj<-coxph(Surv(time,outcome)~hormonfac+age+meno+size+
               as.factor(grade)+enodes+pgr+er+year_cat+chemo,
               data=rott,x=TRUE)

adjsurv <- adjustedsurv(data=rott,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adj,
                        conf_int=TRUE,
                        bootstrap=F,times=5)
```

SOLUTIONS

```
#It turns out that `hormon` needs to be a factor variable for this
rott$hormonfac<-as.factor(rott$hormon)

#Refit Cox model with hormon as factor
#Also need to use x=TRUE here
cox.adjfac<-coxph(Surv(time,outcome)~hormonfac+age+meno+size+
                  as.factor(grade)+enodes+pgr+er+year_cat+chemo,
                  data=rott,x=TRUE)

adjsurv <- adjustedsurv(data=rott,
                        variable="hormonfac",
                        ev_time="time",
```

```

        event="outcome",
        method="direct",
        outcome_model=cox.adjfac,
        conf_int=TRUE,
        bootstrap=F, times=5)

#risks at time 5
adjsurv$ate_object$meanRisk

```

	estimator	time	treatment	estimate	se	lower	upper
	<char>	<num>	<fctr>	<num>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4480378	0.009598346	0.4292254	0.4668502
2:	GFORMULA	5	1	0.3639846	0.020989099	0.3228467	0.4051225

```

#risk diff at time 5
adjsurv$ate_object$diffRisk

```

	estimator	time	A	estimate.A	B	estimate.B	estimate	se
	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4480378	1	0.3639846	-0.08405317	0.022297
	lower		upper	p.value				
	<num>		<num>	<num>				
1:	-0.1277545	-0.04035185	0.0001634392					

```

#risk ratio at time 5
adjsurv$ate_object$ratioRisk

```

	estimator	time	A	estimate.A	B	estimate.B	estimate	se
	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4480378	1	0.3639846	0.8123971	0.04851585
	lower		upper	p.value				
	<num>		<num>	<num>				
1:	0.7173078	0.9074865	0.0001102563					

The estimates are shown in the table below. The 95% CIs have been obtained using an approximation. As an alternative, bootstrap CIs can be obtained using `bootstrap=T, n_boot=1000` in `adjustedsurv` (for 1000 bootstraps) - this is much slower.

Estimand	Estimate	95% CI
Marginal risk up to 5 years: hormon=1	0.36	0.32 to 0.41
Marginal risk up to 5 years: hormon=0	0.45	0.43 to 0.47
Marginal risk difference	-0.084	-0.127 to -0.040
Marginal risk ratio	0.81	0.72 to 0.91

Question 6

So far we have focused on estimating the effect of hormone therapy on the risk up to time 5. We now extend this to obtain estimates of the marginal adjusted survival curves under the two treatment strategies.

- Use the code below to do this using `adjustedsurv`. The solution file shows how to do this without using `adjustedsurv`.
- What is the interpretation of the resulting curves?

```
adjsurv <- adjustedsurv(data=roth,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adjfac,
                        conf_int=TRUE,
                        bootstrap=F)

plot(adjsurv, conf_int=T, custom_colors=c("black", "grey"),
     xlab="Time (years)",
     legend.title="Hormone therapy", ylim=c(0,1))
```

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```
adjsurv <- adjustedsurv(data=roth,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adjfac,
                        conf_int=TRUE,
```

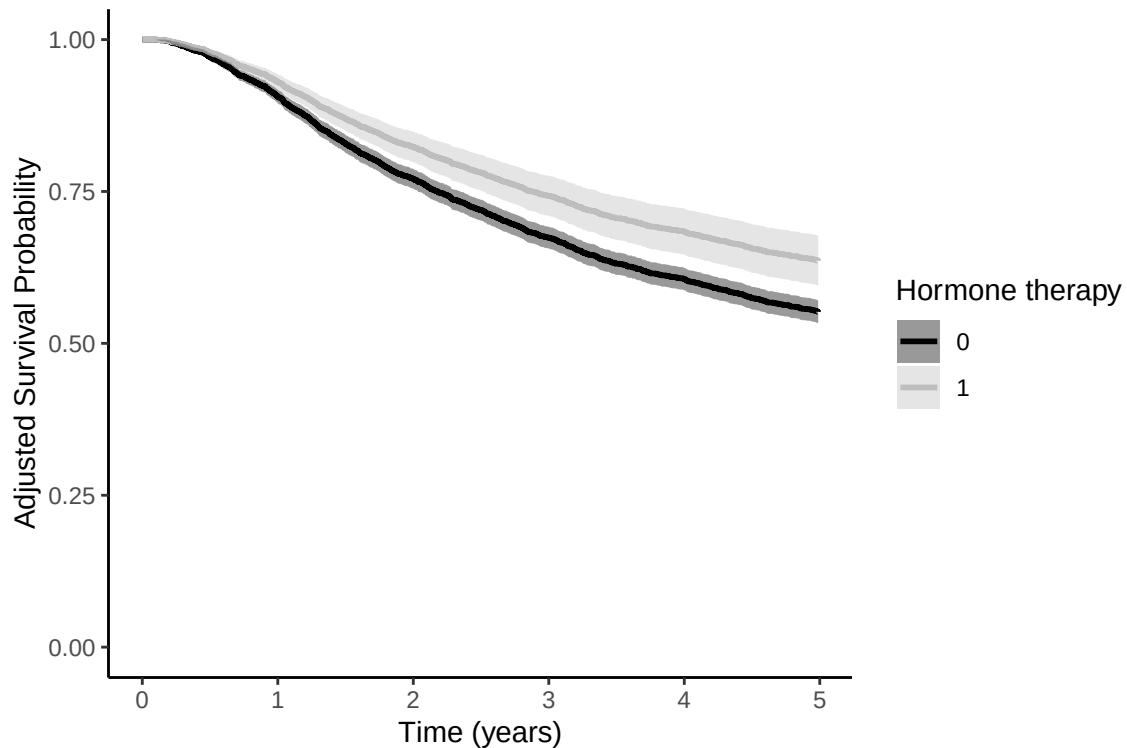
```

bootstrap=F)

plot(adjsurv,conf_int=T,custom_colors=c("black","grey"),
     xlab="Time (years)",
     legend.title="Hormone therapy",ylim=c(0,1))

```

Loading required namespace: pammttools



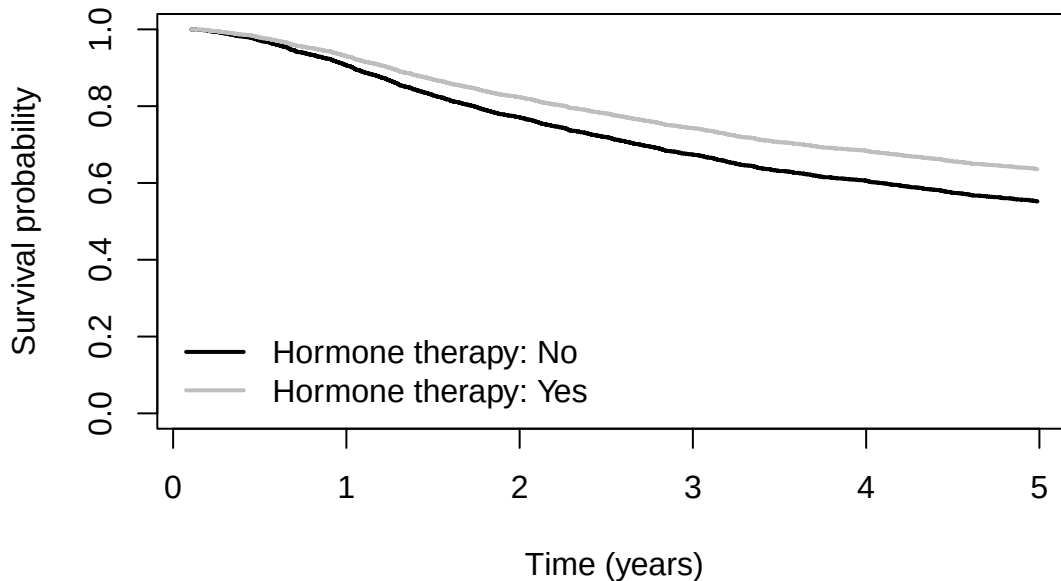
```

#---
# 'by hand' implementation
# predicted survival probabilities at each observed event time
# for each individual, based on cox.adj
event.times<-unique(sort(rott$time[rott$outcome==1]))
surv.hormon1<-predictSurvProb(cox.adj,newdata=data.hormon1,times=event.times)
surv.hormon0<-predictSurvProb(cox.adj,newdata=data.hormon0,times=event.times)

# mean survival probability at each time, averaging over all individuals
survmean.hormon1<-colMeans(surv.hormon1)
survmean.hormon0<-colMeans(surv.hormon0)

```

```
#mean survival curves under the two treatment strategies in a plot
plot(event.times,survmean.hormon0,
     type="s",col="black",lwd=2,ylim=c(0,1),
     xlab="Time (years)",ylab="Survival probability")
lines(event.times,survmean.hormon1,lwd=2,type="s",col="grey")
legend(x="bottomleft",
     c("Hormone therapy: No","Hormone therapy: Yes"),
     col=c("black","grey"),lty=1,lwd=2,bty="n")
```



The curves are the estimated survival curves (for recurrence-free survival) for this population if everyone had used hormone therapy and if everyone had not used hormone therapy.

Extra questions/examples

Question 7

In the above analyses we obtained marginal risks (and risk differences and risk ratios) based on standardisation of conditional estimated from the Cox model `cox.adj` (or equivalently `cox.adjfac`). This made the proportional hazards assumption for all variables included in the model.

In this question we relax the proportional hazards assumption for `hormon` by stratifying on this variable, i.e. by allowing a separate baseline hazard in the `hormon=0` and `hormon=1` groups. After fitting this modified model, the subsequent standardisation steps are the same as above.

See the code below for how to do this, and compare the results with those from Question 5.

```
#Refit Cox model stratified by hormone status
cox.adjstrat<-coxph(Surv(time,outcome)~strata(hormonfac)+age+meno+size+
  as.factor(grade)+enodes+pgr+er+year_cat+chemo,
  data=roth,x=TRUE)

adjsurv <- adjustedsurv(data=roth,
  variable="hormonfac",
  ev_time="time",
  event="outcome",
  method="direct",
  outcome_model=cox.adjstrat,
  conf_int=TRUE,
  bootstrap=F,times=5)

#risks at time 5
adjsurv$ate_object$meanRisk

#risk diff at time 5
adjsurv$ate_object$diffRisk

#risk ratio at time 5
adjsurv$ate_object$ratioRisk
```

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```
#Refit Cox model stratified by hormone status
cox.adjstrat<-coxph(Surv(time,outcome)~strata(hormonfac)+age+meno+size+
  as.factor(grade)+enodes+pgr+er+year_cat+chemo,
  data=roth,x=TRUE)

adjsurv <- adjustedsurv(data=roth,
  variable="hormonfac",
  ev_time="time",
  event="outcome",
  method="direct",
  outcome_model=cox.adjstrat,
  conf_int=TRUE,
  bootstrap=F,times=5)
```

```
#risks at time 5
adjsurv$ate_object$meanRisk
```

	estimator	time	treatment	estimate	se	lower	upper
	<char>	<num>	<fctr>	<num>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4475301	0.009775081	0.4283712	0.4666889
2:	GFORMULA	5	1	0.3669949	0.023264124	0.3213980	0.4125917

```
#risk diff at time 5
adjsurv$ate_object$diffRisk
```

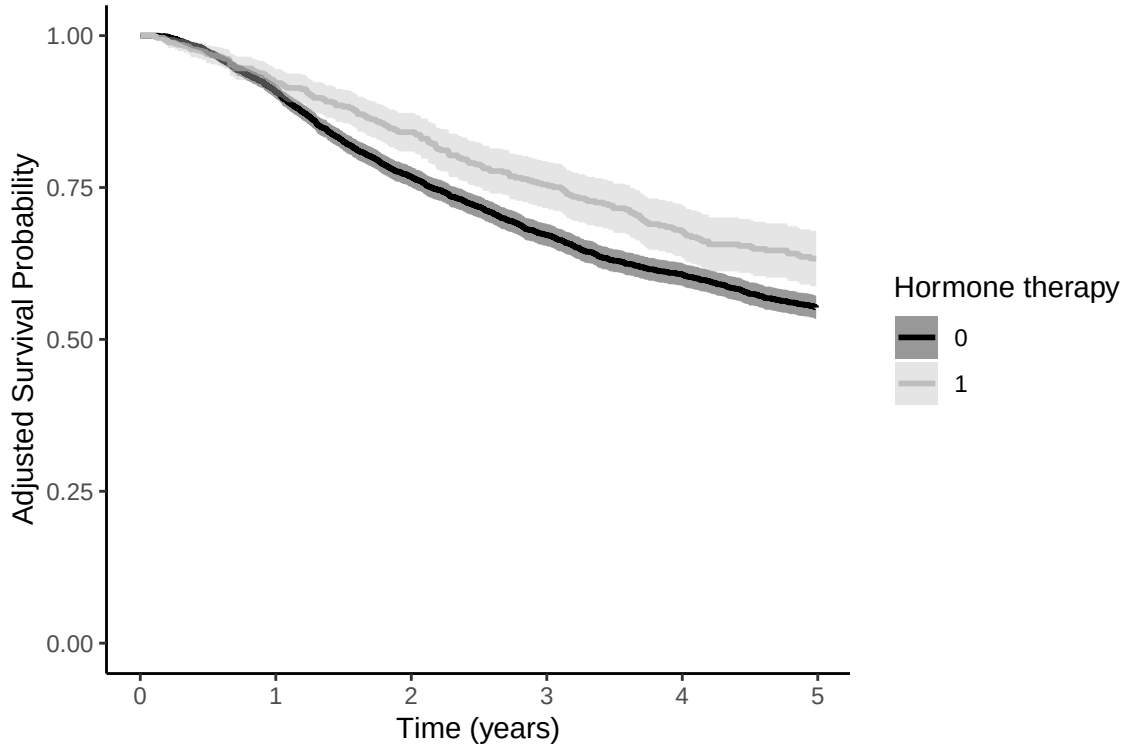
	estimator	time	A	estimate.A	B	estimate.B	estimate	se
	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4475301	1	0.3669949	-0.0805352	0.02523122
			lower	upper			p.value	
			<num>	<num>			<num>	
1:			-0.1299875	-0.03108292			0.001413466	

```
#risk ratio at time 5
adjsurv$ate_object$ratioRisk
```

	estimator	time	A	estimate.A	B	estimate.B	estimate	se
	<char>	<num>	<char>	<num>	<char>	<num>	<num>	<num>
1:	GFORMULA	5	0	0.4475301	1	0.3669949	0.8200452	0.05497686
			lower	upper			p.value	
			<num>	<num>			<num>	
1:			0.7122925	0.9277978			0.001063059	

```
#---
#We can also obtain survival curve plots
adjsurv.plot <- adjustedsurv(data=roth,
                             variable="hormonfac",
                             ev_time="time",
                             event="outcome",
                             method="direct",
                             outcome_model=cox.adjstrat,
                             conf_int=TRUE,
                             bootstrap=F)
```

```
plot(adjsurv.plot,conf_int=T,custom_colors=c("black","grey"),
     xlab="Time (years)",
     legend.title="Hormone therapy",ylim=c(0,1))
```



The estimates are shown in the table below. The results based on the stratified Cox model are very similar to those based on the model assuming proportionality of hazards for hormone therapy.

Estimand	Estimate	95% CI
Cox model assuming proportional hazards for hormon		
Marginal risk up to 5 years: hormon=1	0.36	0.32 to 0.41
Marginal risk up to 5 years: hormon=0	0.45	0.43 to 0.47
Marginal risk difference	-0.084	-0.127 to -0.040
Marginal risk ratio	0.81	0.72 to 0.91
Cox model stratified by hormon		
Marginal risk up to 5 years: hormon=1	0.37	0.32 to 0.41
Marginal risk up to 5 years: hormon=0	0.45	0.43 to 0.47
Marginal risk difference	-0.081	-0.130 to -0.031

Estimand	Estimate	95% CI
Marginal risk ratio	0.82	0.71 to 0.93

Question 8

Another estimand of interest could be the restricted mean survival times (RMST) under the two treatment strategies, or differences in the RMST. Based on the estimates of the marginal risks (or survival probabilities) obtained above, here we show how to estimate the RMST up to time 5 under the two treatment strategies, and their difference.

The code provided illustrates how to estimate the marginal RMST and RMST differences using `adjustedsurv`.

Interpret the RMST estimates and their difference.

```
#---
#Obtain RMST under each treatment strategy using adjustedCurves.
#Bootstrapping is required to get the confidence intervals for the RMST.
#Note you should use at least 500 bootstrap samples,
#but this is quite slow so we have just used 50 below for illustration.

adjsurv <- adjustedsurv(data=roth,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adjf,
                        conf_int=TRUE,
                        bootstrap=T,n_boot=50)

#RMST under the two treatment strategies
adjrmst <- adjusted_rmst(adjsurv,
                        from=0, to=max(roth$time[roth$outcome==1]),
                        conf_int=TRUE,conf_level=0.95)

adjrmst

#RMST difference
adjrmst.diff <- adjusted_rmst(adjsurv,
                             from=0, to=max(roth$time[roth$outcome==1]),
                             conf_int=TRUE,conf_level=0.95,
                             contrast = "diff")

adjrmst.diff
```

SOLUTIONS

```
#---
#Obtain RMST under each treatment strategy using adjustedCurves.
#Bootstrapping is required to get the confidence intervals for the RMST.
#Note you should use at least 500 bootstrap samples,
#but this is quite slow so we have just used 50 below for illustration.

adjsurv <- adjustedsurv(data=roth,
                        variable="hormonfac",
                        ev_time="time",
                        event="outcome",
                        method="direct",
                        outcome_model=cox.adfac,
                        conf_int=TRUE,
                        bootstrap=T,n_boot=50)

#RMST under the two treatment strategies
adjrmst <- adjusted_rmst(adjsurv,
                        from=0, to=max(roth$time[roth$outcome==1]),
                        conf_int=TRUE,conf_level=0.95)

adjrmst
```

	to	group	rmst	se	ci_lower	ci_upper	n_boot
1	4.988364	0	3.721346	0.03221870	3.658198	3.784493	50
2	4.988364	1	3.986890	0.05967903	3.869921	4.103859	50

```
#RMST difference
adjrmst.diff <- adjusted_rmst(adjsurv,
                              from=0, to=max(roth$time[roth$outcome==1]),
                              conf_int=TRUE,conf_level=0.95,
                              contrast = "diff")

adjrmst.diff
```

	to	diff	se	ci_lower	ci_upper	p_value
1	4.988364	-0.2655445	0.06782058	-0.3984704	-0.1326186	9.02554e-05

The RMST is interpreted as the expected time alive and free of disease recurrence up to 5 years. The estimated RMST if everyone were to use hormone therapy is 3.99 years, and estimated RMST if everyone were not to use hormone therapy is 3.72 years. The RMST

difference between the two treatment strategies is 0.27 (95% CI 0.14 to 0.39). This means that if everyone had not used hormone therapy, the expected time alive and free of disease recurrence up to 5 years would be 0.27 years less than if everyone had used hormone therapy.

Question 9

To estimate marginal risks using standardisation based on fully parametric models, such as a Weibull, you can use the `mexhaz` package, as shown below.

```
#fit the weibull model
mex.weibull<-mexhaz(formula=Surv(time,outcome)~hormon+age+meno+size+
                    as.factor(grade)+enodes+pgr+er+year_cat,
                    data=rott, expected = NULL, base = "weibull")

#obtain marginal survival probabilities at time 5, and their difference
adjsurv.weibull <- adjsurv(mex.weibull,time.pts=5,
                          data=data.hormon1,data.0=data.hormon0)

adjsurv.weibull

#adj.surv is Pr(T>5|do(X=1)), (adj.surv.inf, adj.surv.sup) is the 95% CI
#adj.surv.0 is Pr(T>5|do(X=0)), (adj.surv.0.inf, adj.surv.0.sup) is the 95% CI
#diff.adj.surv is Pr(T>5|do(X=1))-Pr(T>5|do(X=0)) (1 minus the risk diff),
# and (diff.adj.surv.inf,diff.adj.surv.sup) is the 95% CI
```

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```
#fit the weibull model
mex.weibull<-mexhaz(formula=Surv(time,outcome)~hormon+age+meno+size+
                    as.factor(grade)+enodes+pgr+er+year_cat,
                    data=rott, expected = NULL, base = "weibull",
                    print.level = 0)
```

Data

Name	N.Obs.Tot	N.Obs	N.Events	N.Clust
rott	2982	2982	1285	1

Details

Iter	Eval	Base	Nb.Leg	Nb.Aghq	Optim	Method	Code	LogLik	Total.Time
------	------	------	--------	---------	-------	--------	------	--------	------------

```
6      9 weibull      20      10      nlm      ---      1 -3758.162      0.07
```

```
#obtain marginal survival probabilities at time 5, and their difference
adjsurv.weibull <- adjsurv(mex.weibull,time.pts=5,
                           data=data.hormon1,data.0=data.hormon0)
adjsurv.weibull$results
```

```
time.pts  adj.surv  adj.surv.inf  adj.surv.sup  adj.surv.0  adj.surv.0.inf
1         5 0.6176605    0.6596288    0.5723863  0.5439169    0.5611985
adj.surv.0.sup  diff.adj.surv  diff.adj.surv.inf  diff.adj.surv.sup
1         0.5262753    0.0737436    0.02680046    0.1206867
```

The estimated marginal risk difference up to 5 years based on the Weibull model is -0.073, with 95% CI -0.121 to -0.027. This compares with the estimate of -0.084 (95% CI -0.127 to -0.040) based on the Cox model.